

Cerebellar Transcranial Magnetic Stimulation (TMS) Impairs Visual Working Memory

Viñas-Guasch et al. (2023) | Cerebellum, 22(3), 332-347

What Was Their Question?

- The cerebellum has traditionally been seen as a motor structure, but evidence has been building that it supports cognition through modality-specific cerebro-cerebellar loops.
- Prior work confirmed two cerebro-cerebellar loops for verbal WM: right superior cerebellum for articulatory rehearsal, right inferior cerebellum for phonological storage.
- Core question: does the same kind of modality specificity exist for visual WM, specifically in the left cerebellum?

How Did They Do It?

FMRI-GUIDED TARGETING

- Participants first did fMRI with a Sternberg Visual WM task (novel snowflake-like geometric patterns — chosen specifically because they can't be verbally rehearsed)
- fMRI maxima within each individual's left superior (lobule VI) and left inferior (lobule VIIIa/b) cerebellum were identified as TMS targets

STUDY 1 — LEFT SUPERIOR CEREBELLUM

- 40 participants. Double-pulse TMS (120% rMT, 1Hz) delivered at end of encoding phase on 50% of trials
- 4 conditions: Visual WM TMS, Visual WM Sham, Motor Control (no WM), Visual Control (no WM)

STUDY 2 — LEFT INFERIOR CEREBELLUM

- 14 participants (subset of Study 1). Same TMS protocol applied to inferior cerebellar target
- 2 tasks compared: Visual WM TMS and Verbal WM TMS — the key test for modality specificity

What Did They Find?

- Study 1 (superior): TMS significantly reduced accuracy in Visual WM (83.87% non-stim vs 79.92% stim, $p=.025$) and slowed response times ($p<.05$). No effect on motor or visual control conditions.
- Study 2 (inferior): TMS significantly reduced accuracy in Visual WM (85.00% non-stim vs 79.38% stim, $p<.05$) but NOT in Verbal WM (82.16% vs 80.15%, $p>.3$).
- Critical comparison: the accuracy drop was larger for inferior cerebellar TMS than superior (paired t-test $p<.05$). The inferior cerebellum is both modality-specific and more impactful.

What Does It Mean?

- The cerebellum is not just a motor controller. It participates in WM through distinct cerebro-cerebellar loops with modality specificity — different subregions handle visual vs verbal content.
- Left inferior cerebellum (lobule VIIIa/b) is specifically necessary for visual WM. The authors propose it supports directing visual attention to target features during encoding and possibly error prediction during maintenance.
- The inferior cerebellum connects to the dorsal attention network — suggesting its WM role may involve attentional allocation to visual features as much as storage itself.

What Are the Problems?

- Study 2 had only 14 participants — the power analysis showed you'd need 36 for adequate power with this design. Results are suggestive, not definitive.
- Only left hemisphere was directly tested for the visual-verbal dissociation. The right inferior cerebellum pilot (n=9) is promising but very small.
- TMS applied at end of encoding phase. We know encoding was affected but can't isolate whether maintenance or retrieval were also disrupted.
- The visual WM task used non-nameable snowflake patterns specifically to prevent verbal rehearsal. This maximizes experimental control but may not generalize to naturalistic visual WM like faces or spatial navigation.

Discussion Questions

- This study shows the cerebellum has modality-specific WM roles most clinicians don't know about. A patient with a cerebellar lesion might have impaired visual WM but intact verbal WM, or vice versa. Why do you think standard neurological exams would miss this?
- The inferior cerebellum connects to the dorsal attention network. Given that, do you think its role in visual WM is more about storing visual information, or about directing attention to the visual features that need to be encoded? How would you test that?
- TMS demonstrates necessity by disruption. fMRI demonstrates correlation by activation. A region being necessary for a task and a region activating during a task are different claims. Which type of evidence do you trust more, and what can't each one tell you?